MILITARY SPECIFICATION SHEET

ELECTRON TUBE, GAS SWITCHING

TYPE 6322

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: TR, narrow band, tunable frequency range 1,215 to 1,355 MHz

ABSOLUTE RATINGS:

Parameter:	Ii	Transmitter po	Ignitor Ebb	Alt
Unit:	μAdc	kw	Vdc	ft
Maximum:	200	450	-1,000	10,000
Minimum:	100	---	-750	---

PHYSICAL CHARACTERISTICS:

Dimensions:	See figure 1
Mounting position:	Any
Weight:	1.5 ounces (approx)

TEST CONDITIONS:

Parameter:	F	po	tp1	tp2	prr	σ'	Ii
Unit:	MHz	kw	μs	μs	pps	---	μAdc
Test 1:	1,285 ±2%	5 ±20%	4	1.0	250	1.15	100
Test 2:	1,285 ±2%	450 ±10%	4	---	250	1.15	100

GENERAL:

Qualification - Required

(E) denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
	Quality conformance inspection, part 1	1	---					
4401	Ignitor ignition time	2	---	Ebb = 675 Vdc; Ri = 2.0 Meg	t	---	5.0	sec
4406	Ignitor voltage drop	2	---	Ii = 100 μAdc	Eid	300	450	Vdc
4426	Tuning range	3, 5	---		F	1,215	1,355	MHz
4416	Insertion loss	3, 11	---	F = 1,285 MHz ±1%	Li	---	0.8	dB
4421	Ignitor interaction	---	---	Iz = 100 μAdc	Li	---	0.1	dB
(E) 4452	Spike leakage energy	6	1		Ws	---	0.35	erg
(E) 4452	Flat leakage power	6	1		pf	---	90	mW
	Quality conformance inspection, part 2							
(E) ---	Ignitor oscillation	---	---		Ii	---	50	μAdc
4461	Loaded Q	3	---	F = 1,285 MHz ± 1%	QL	175	250	---
4471	Recovery time	7	2		t	---	25	μs
4466	Temperature coefficient of frequency	3	---	F = 1,285 MHz ± 1%	ΔF	0	-10	MHz
1027	Temperature cycling life test	3, 8	---	-40° to +100°C	---	12	---	Cycles
	Quality conformance inspection, part 3							
---	Life-test provisions	7, 10	2	Group D	t	500	---	hrs
---	Life-test end points	---	---					
4406	Ignitor voltage drop	---	---	Ii = 100 μAdc	Eid	---	500	Vdc
4421	Ignitor interaction	---	---	Iz = 100 μAdc	Li	---	0.3	dB
4416	Insertion loss	3, 11	---	F = 1,285 MHz ± 1%	Li	---	1.0	dB
4471	Recovery time	7	2		t	---	40	μs
4452	Spike leakage energy	6	1		Ws	---	0.5	erg
	Periodic-check tests							
4021	Degradation due to vibration	3, 4	---	F = 1,285 MHz ± 1%	---	---	---	---
4426	Tuning range	3, 5, 9	---		F	1,215	1,355	MHz

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NOTES:

1. All tests listed under quality conformance inspection, part 1, shall have an AQL of 1.0 percent and an inspection level II.
2. This test to be performed at the conclusion of the holding period.
3. The tube shall be mounted in a cavity as shown in figure 3.
4. The tube and cavity shall be vibrated in the direction of the keep-alive axis for 12 hours at 2.0 G with an amplitude of motion of 0.040 inch. After this, the tuning shall not have changed more than 1.0 MHz from its initial value and the tube shall satisfy all other electrical tests of this specification.
5. The tuning screw shall turn one-half turn from its stop position before the 1,215 MHz position is reached. A minimum of seven turns of the tuning screw shall be required to cover the specified tuning range. The cones shall not touch when the tuning knob is tuned through its entire range (as shown by an electrical continuity test).
6. The tube shall be mounted as shown in figure 3, and tested as shown in figure 2. The loaded Q of the cavity shall be between 175 and 250, with a matched termination on the output of the cavity. A phase shifter shall be inserted in the test line preceding the cavity and adjusted for maximum leakage at each plus width.
7. The tube shall be mounted in a cavity as shown in figure 4. The loaded Q shall be within 120 and 140, with a matched termination on the output and the cavity shunt mounted on the waveguide.
8. (E) The tube shall be subjected to the number of temperature cycles and temperature limits specified. The tube shall be maintained at each end-point temperature for sufficient time to reach equilibrium, but not for less than 5 minutes. Changes in temperature from end point to end point shall be gradual but the cycle shall not exceed 30 minutes. After cycling, the tube shall be allowed to return to room temperature, held for 24 hours, and then shall be tested for the specified limits of ignitor ignition time and ignitor voltage drop.
9. The tuner shall be cycled through the specified tuning range a minimum of 10 times before the tuning range is checked.
10. The tube shall be mounted as specified for the recovery time measurement, except that the output of the cavity will be terminated by a 50-ohm matched load.
11. The cavity shown in figure 3 shall be calibrated, as required, using the following procedure. The calibrator shown in figure 5 shall be inserted into the cavity and tuned to resonance. The calibrator shall be removed and the TUT shall be inserted into the cavity and tuned to resonance. The insertion loss is the difference between the readings obtained with the calibrator and tube in the cavity.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army - EL
Air Force - 11, 17, 80
DSA - ES

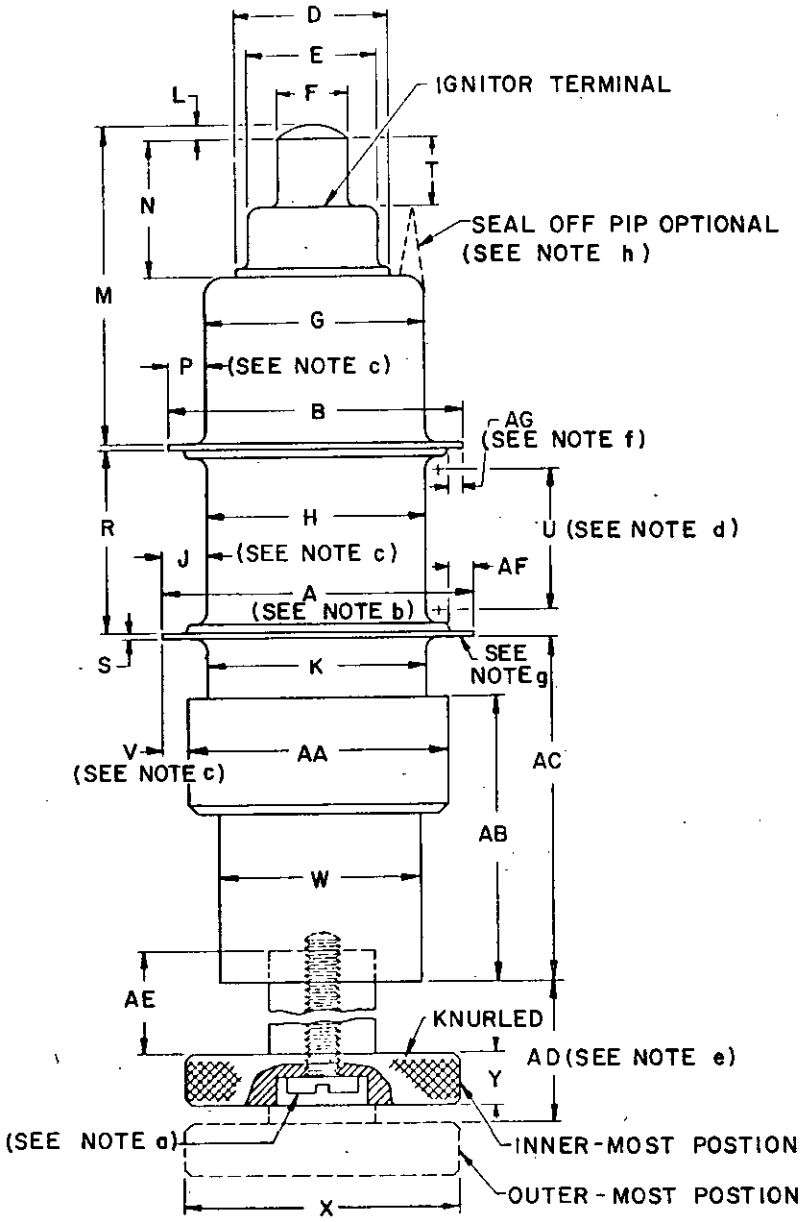
User activities:
Navy - AS, OS, MC, CG, SH

Preparing activity: Navy - EC

Agent: DSA - ES

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NOTES:

a. The tuning knob may be detached by removing screw. A screw driver slot (.391 (9.93 mm) max long, .047 (1.19 mm) max wide, .040 ±.002 (1.02 ±.05 mm) deep) is provided in the head of the tuning screw which is exposed when the tuning knob is removed.
b. Contact discs shall be concentric within .016 (.41 mm).
c. Applies at all points around tube except fillets formed at seal with contact discs.
d. Locate approximately centrally between contact discs.
e. Applies when tuning screw is in its extreme position.
f. Contact portion of disc measured radially must be free from splits or tears and must be smooth and flat to the extent controlled by dimension R.
g. Contact discs shall be gold plated (10 - msi min) or silver plated (15 - msi min) on external surfaces.
h. Seal-off pip shall not extend above broken line..

FIGURE 1. Outline drawing of electron tube type 6322.

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection part 1 (see note)		
A	1.054 (26.77)	1.070 (27.18)
B	.995 (25.27)	1.005 (25.53)
M	1.125 (28.58)	1.250 (31.75)
R	.670 (17.02)	.675 (17.15)
AC	1.250 (31.75)	1.375 (34.93)
Quality conformance inspection, part 2		
J	.125 (3.18)	
K	.719 (18.26)	.781 (19.84)
P	.093 (2.36)	
S		.010 (.25)
U	.438 (11.13) Nom	
V	.070 (1.78)	
X		.938 (23.83)
AB		1.094 (27.79)
AF	.070 (1.78)	
AG	.040 (1.02)	
AD		.500 (12.70)
Quality conformance inspection, part 3 (periodic check tests)		
D	.500 (12.70) nom	
E	.426 (10.82) nom	
F	.248 (6.30)	.252 (6.40)
G	.719 (18.26)	.781 (19.84)
H	.719 (18.26)	.781 (19.84)
L	.031 (.79)	
N	.489 (12.42) nom	
T	.234 (5.94)	.266 (6.76)
W	.672 (17.07) nom	
Y	.188 (4.78) nom	
AA		.875 (22.23)
AE	.234 (5.94)	.266 (6.76)

NOTE: Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1 shall be 1.0 percent, inspection level II.

FIGURE 1. Outline drawing of electron tube type 6322 - continued.

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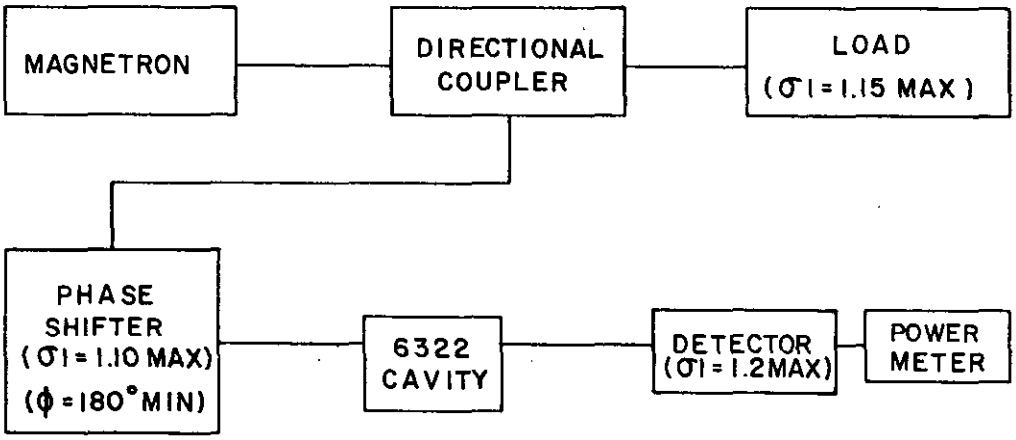


FIGURE 2. Test circuit.

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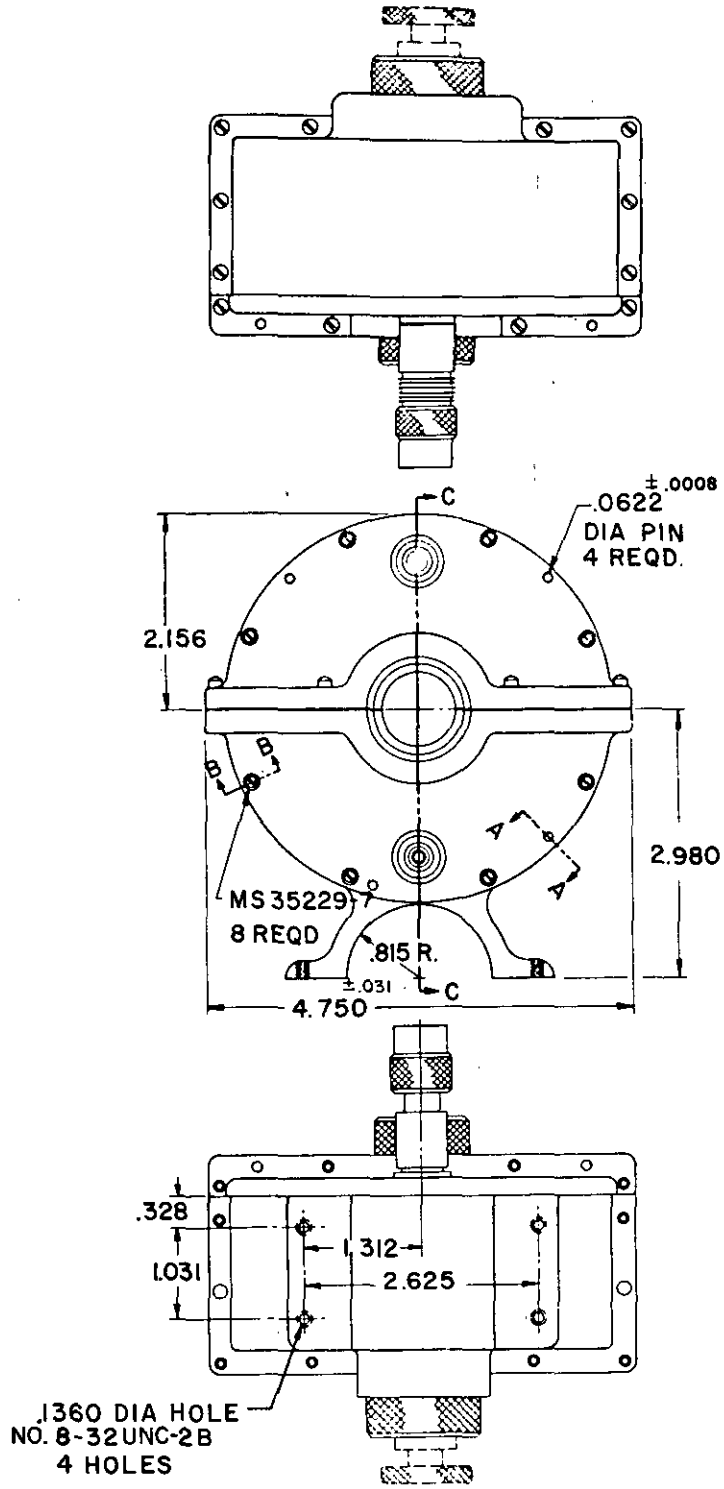
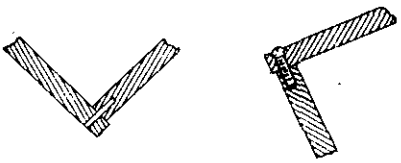


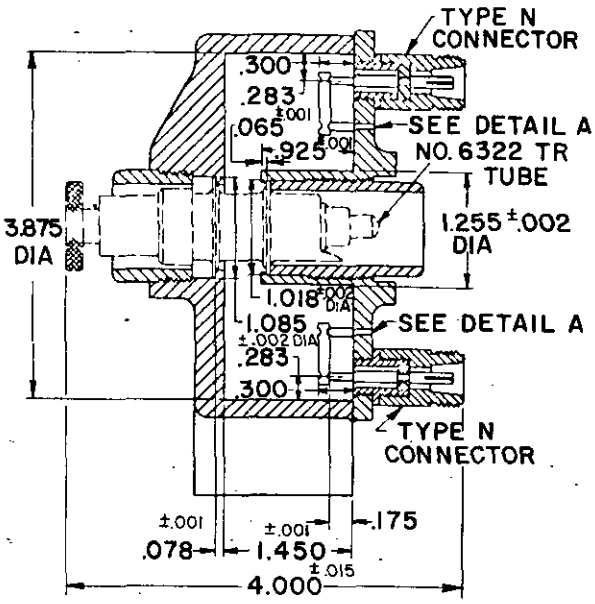
FIGURE 3. Tuning cavity A.

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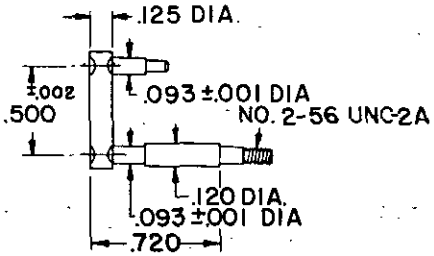


SECTION A-A SECTION B-B

INCHES	MM	INCHES	MM
.0008	.02	.500	12.70
.001	.03	.720	18.29
.002	.05	.815	20.70
.016	.41	.925	23.50
.031	.79	1.018	25.86
.0622	1.58	1.031	26.19
.065	1.65	1.085	27.56
.078	1.98	1.255	31.88
.093	2.36	1.312	33.32
.120	3.05	1.450	36.83
.125	3.18	2.156	54.76
.1360	3.45	2.625	66.68
.175	4.45	2.980	75.69
.283	7.19	3.875	98.43
.300	7.62	4.000	101.60
.328	8.33	4.750	120.65



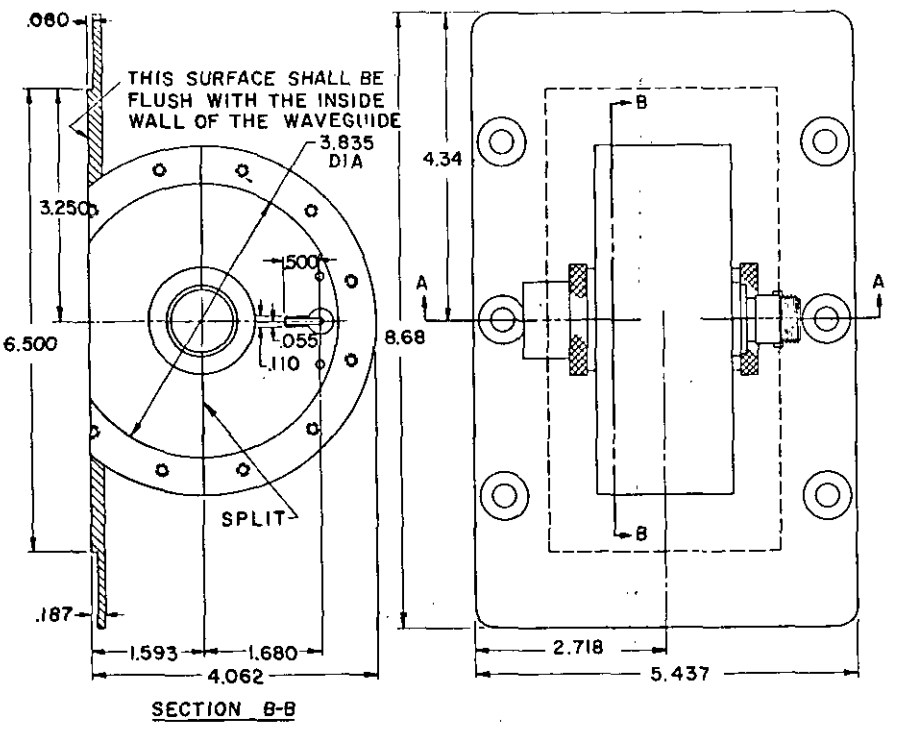
SECTION C-C

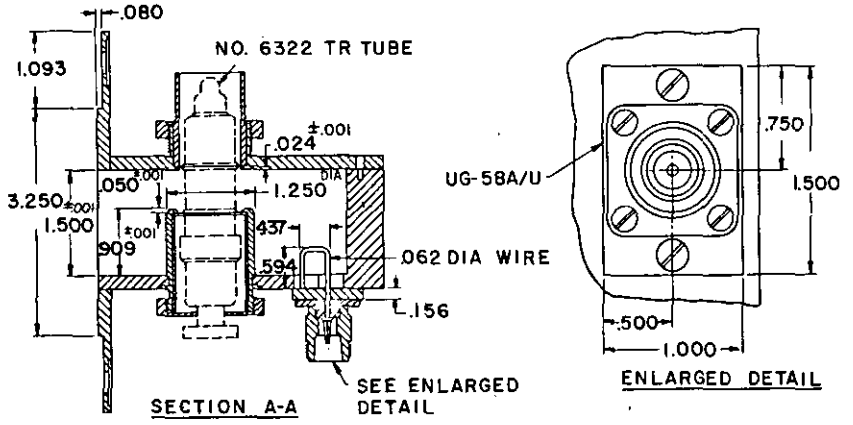


DETAIL A

FIGURE 3. Tuning cavity A - continued.

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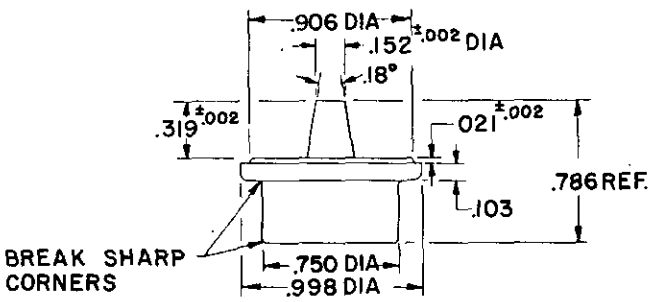


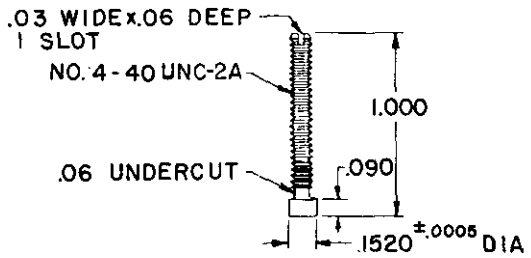


INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.001	.03	.080	2.03	.438	11.13	1.000	25.40	1.680	42.67	4.344	110.34
.024	.61	.083	2.11	.500	12.70	1.094	27.79	2.719	69.06	5.438	138.13
.050	1.27	.110	2.79	.594	15.09	1.250	31.75	3.250	82.55	6.500	165.10
.055	1.40	.156	3.96	.750	19.05	1.500	38.10	3.835	97.41	8.688	220.68
.063	1.60	.187	4.75	.909	23.09	1.593	40.46	4.063	103.20		

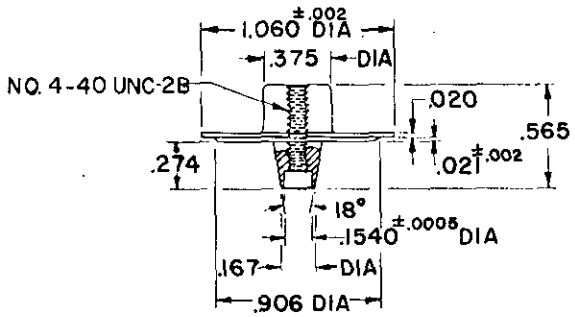
FIGURE 4. Tuning cavity B.

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INCHES	MM
.0005	.01
.002	.05
.020	.51
.021	.53
.063	1.60
.090	2.29
.103	2.62
.125	3.18
.152	3.86
.1520	3.86
.1540	3.91
.167	4.24
.274	6.96
.319	8.10
.375	9.53
.565	14.35
.750	19.05
.786	19.96
.906	23.01
.998	25.35
1.000	25.40
1.060	26.92



NOTES:
a. Unless otherwise specified, tolerances are ±.005 (.13 mm) on decimals and ½° on angles.
b. Material, brass-silver plate, 1 required of each.

FIGURE 5. Cavity A calibrator.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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